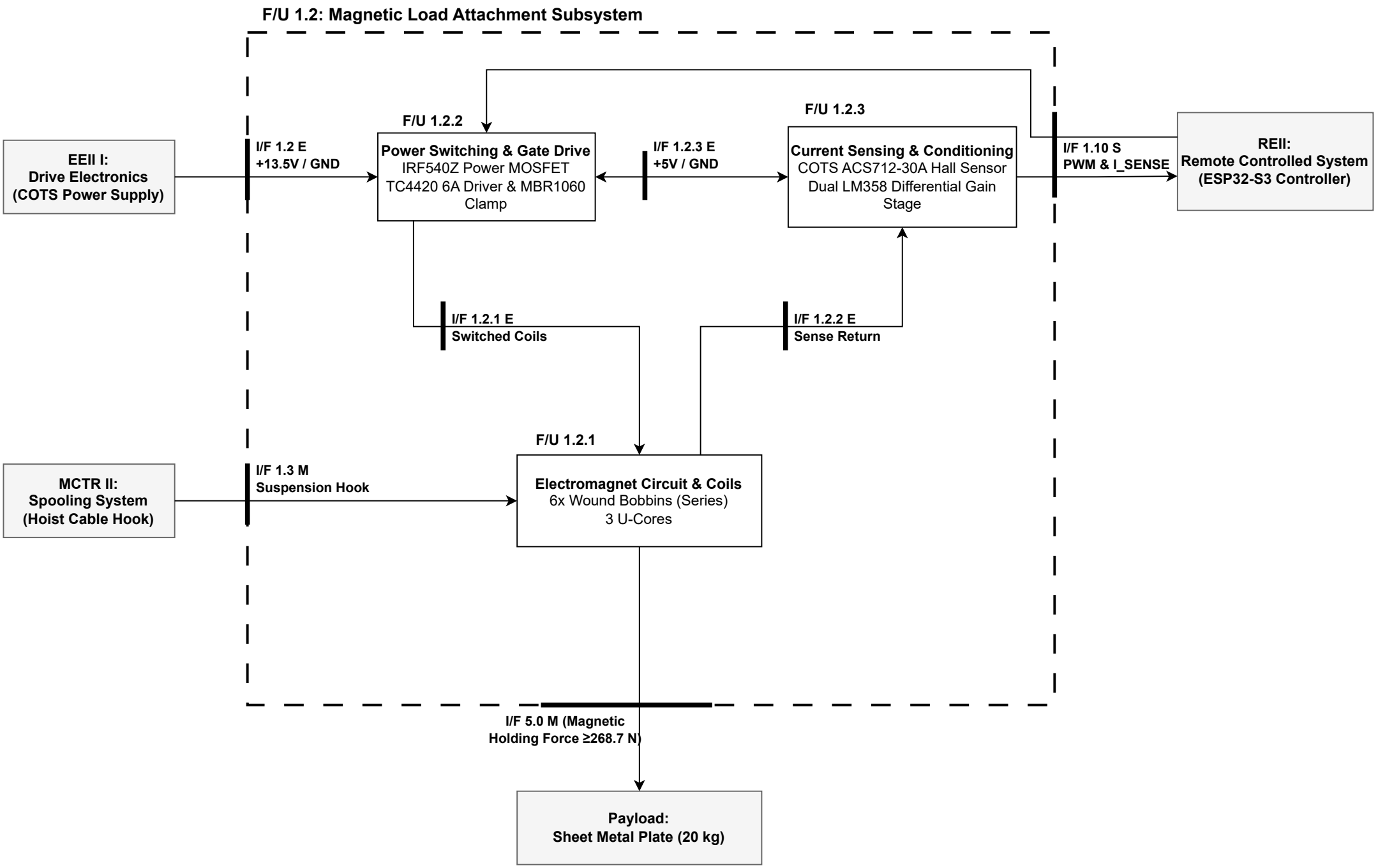

F/U 1.2: Magnetic Load Attachment Subsystem
EEll I:
Drive Electronics
(COTS Power Supply)
I/F 1.2 E
+13.5V / GND
F/U 1.2.2
Power Switching & Gate Drive
IRF540Z Power MOSFET
TC4420 6A Driver & MBR1060
Clamp
I/F 1.2.3 E
+5V / GND
F/U 1.2.3
Current Sensing & Conditioning
COTS ACS712-30A Hall Sensor
Dual LM358 Differential Gain
Stage
I/F 1.10 S
PWM & I_SENSE
REll:
Remote Controlled System
(ESP32-S3 Controller)
I/F 1.2.1 E
Switched Coils
I/F 1.2.2 E
Sense Return
MCTR ll:
Spooling System
(Hoist Cable Hook)
I/F 1.3 M
Suspension Hook
F/U 1.2.1
Electromagnet Circuit & Coils
6x Wound Bobbins (Series)
3 U-Cores
I/F 5.0 M (Magnetic
Holding Force ≥268.7 N)
Payload:
Sheet Metal Plate (20 kg)